Year 9 Algebra Revision — Applications

Harder numbers, less scaffolding, and real-world problems. Show every step of your working.

1 Harder Equations

Expand brackets (watch the signs!) and collect like terms before solving.

1. $3(x+4) = 2(x+9)$ _____

2. $5(x+1) = 2(x+7)$ _____

3. $20 - 3(x-2) = 11$ _____

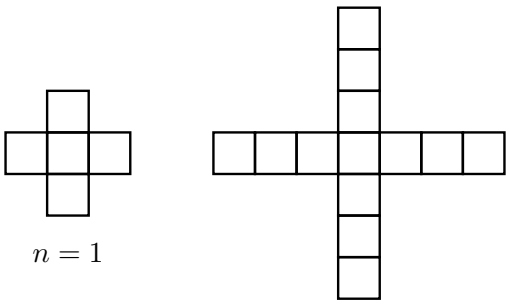
4. $15 - 2(x-1) = x+5$ _____

5. $\dfrac{x+5}{3} = 6$ _____

6. $\dfrac{2x-4}{3} = 4$ _____

2 A Harder Pattern

Crosses are made from square tiles. Only $n = 1$ and $n = 3$ are shown below — $n = 2$ is missing.



$n = 1$

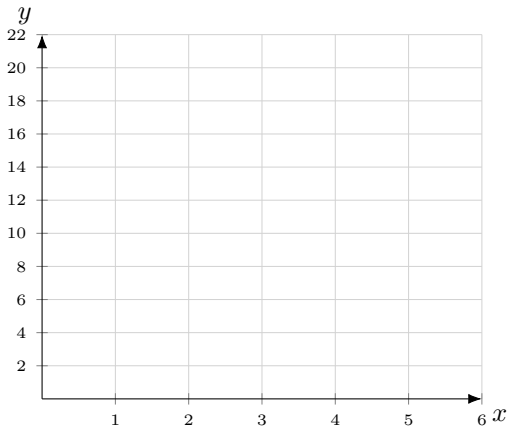
$n = 3$

n	1	2	3	4	5
tiles	5		13		

(a) Using only the $n = 1$ and $n = 3$ diagrams, how many tiles are added each time n increases by 1? _____

(b) Complete the table above.

(c) Write the rule: tiles = _____



Plot the points from your table and draw the line. Gradient: _____ y-intercept: _____

3 Application Problems

1. **Phone plan.** A mobile phone plan costs \$20 per month plus \$0.40 per minute of calls. The total monthly cost, C dollars, for m minutes is

$$C = 0.4m + 20.$$

   (a) Find the cost for a month with 50 minutes of calls. _____

   (b) Jess's bill was \$52. How many minutes did she use? _____

2. **Filling a tank.** A water tank already contains 30 L when a hose starts filling it at a constant rate of 6 L per minute.

   (a) Write an equation for the volume V litres in the tank after t minutes. _____

   (b) Complete the table below using your equation.

t (min)	0	2	4	6	8
V (L)					

   (c) The tank's capacity is 90 L. Use your equation to find how long it takes to fill completely. _____

3. **Taxi fares.** A taxi company charges a fixed flag-fall fee plus a rate per kilometre. The table shows the fare for different distances.

distance, d (km)	2	5	8
fare, F (\$)	9	15	21

   (a) Find the rate per km (the gradient). _____

   (b) Find the flag-fall fee (the y-intercept). _____

   (c) Write the equation for the fare: $F =$ _____

   (d) Use your equation to predict the fare for a 12 km trip. _____

4. **Savings.** Maria is saving money each week. After 3 weeks she has \$85. After 7 weeks she has \$145.

   (a) Find how much she saves each week (the gradient). _____

   (b) Find how much she started with (the y-intercept). _____

   (c) Write the equation for her savings, S, after w weeks: $S =$ _____

   (d) How much will she have after 12 weeks? _____

5. **Comparing two gyms.** Gym A charges a \$50 joining fee plus \$10 per session. Gym B charges no joining fee but \$18 per session.

   (a) Write an equation for the total cost at Gym A after s sessions. _____

   (b) Write an equation for the total cost at Gym B after s sessions. _____

   (c) Find the total cost at each gym after 6 sessions. Which gym is cheaper? _____

   (d) Find the total cost at each gym after 10 sessions. Which gym is cheaper now? _____